Adaptive Preconditioned Stochastic Optimisation for Large-Scale Logistic Regression

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Abstract

Existing optimisation methods do not perform well on large problems that are becoming more relevant practically. An efficient stochastic optimisation method in large-scale logistic regression is proposed. The algorithm uses Stochastic Variance Reduced Gradient Method (SVRG), and a cheap preconditioner based on random sketching of the data. At prescribed intervals, we build a small sketched approximation of the data-weighted Gram matrix (the GLM curvature) and use it to precondition following gradient updates. Iteration costs $O(\text{nnz}(\mathcal{B})) + O(d)$ ($\mathcal{B}$ is a minibatch, d is the parameter dimension) and is fully parallelisable. In the standard assumptions of smoothness and strong convexity through ℓ_2 regularization, the method achieves global linear convergence to an ε-optimal solution. Results demonstrate faster convergence in terms of number of iterations than first-order methods, and lower cost per iteration on sparse high-dimensional data compared to Newton's method.

Keywords: Stochastic optimization, Variance reduction, Preconditioning, Randomized sketching, Logistic regression, Fisher information, Large-scale learning

1 Introduction

Logistic Regression

Logistic regression is a canonical convex optimization problem in machine learning, widely used to model the posterior probabilities of the classes [5]. It involves the minimization of

$$f(\theta) = \frac{\lambda}{2}\|\theta\|^2 + \frac{1}{n}\sum_{i=1}^{n} \ell(b_i a_i^\top \theta), \tag{1}$$

where the logistic loss function is defined as

$$\ell(z) = \log(1 + e^{-z}), \quad z \in \mathbb{R} \tag{2}$$

b_i denotes the class label for the ith sample with feature vector $a_i \in \mathbb{R}^d$. The regulariser $\frac{\lambda}{2}\|\theta\|^2$ with $\lambda > 0$ ensures strong convexity. Each term $\ell(b_i a_i^\top \theta)$ is convex and smooth in θ.

Large-Scale Logistic Regression

Let d denote the dimension of the parameter vector, i.e. the number of features (often denoted as p in statistics), and n the number of data points. In large-scale settings, that is, either $n \gg d$ or d large and sparse, the best optimisation method depends on the computational complexity. Full-batch gradient descent, as a first-order method, has cost $O(nd)$ per step. It is able to achieve linear convergence $O((1 - \frac{\mu}{L})^k)$ under L-smoothness and μ-strong convexity, but typically requires many passes over the data. Newton's Method converges in far fewer iterations due to curvature information, but each step requires forming and inverting the Hessian $H(\theta) = X^\top W(\theta) X + \lambda I$, where $W(\theta)$ is the diagonal weight matrix of second derivatives, costing $O(nd^2)$ or $O(d^3)$ in general. Thus Newton's Method is impractical in large-scale settings discussed here.

Stochastic gradient methods (SGD) use $O(d)$ work per minibatch. However, standard SGD only has a sublinear $O(\frac{1}{k})$ convergence rate for strongly convex sums, which is much slower than gradient descent. Variance-reduced methods such as SAG/SAGA and SVRG combine the low iteration cost of SGD with the fast convergence of batch methods. They achieve linear convergence by storing past gradients or using periodically updated full gradients. That being said, those methods still can suffer from ill-condit ioning. Their number of iterations increases as the condition number $\frac{L}{\mu}$ increases. Accelerated gradient methods

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(Nesterov's AGD) improve this to $O(\sqrt{\frac{L}{\mu}}\log(\frac{1}{\varepsilon}))$ iterations, but each iteration is still $O(nd)$, or $O(nnz)$ (often $O(d)$) in high-dimensional sparse settings, and acceleration is sensitive to noise in stochastic gradients.

An alternative is to introduce curvature information. Quasi-Newton and L-BFGS methods maintain Hessian approximations, but require $O(d^2)$ storage or operations, and are not easily parallelised on very large data. Subsampled or sketched Newton methods approximate the Hessian from a random data subset, achieving faster convergence, but they still perform (approximate) second-order updates and may break sparsity. Coordinate and block-coordinate methods exploit sparsity but do not fully leverage curvature and may converge slowly if features are correlated.

Related Work

Our work draws on several strands of optimization research. Specifically, we wish to combine first-order efficiency and second-order convergence speeds. We briefly highlight the most relevant.

Gradient descent (GD) for L-smooth, μ-strongly convex functions converges at a linear rate $(1-\frac{\mu}{L})^k$. Nesterov's accelerated gradient descent [8] improves this to $O\big(\sqrt{\frac{L}{\mu}},\log(\frac{1}{\varepsilon})\big)$ iterations, effectively square-rooting the condition number. However, AGD still requires full-gradient computations or reliable gradient information. Applying it in a stochastic context can be delicate. Recent work also includes accelerated stochastic gradient or optimal finite-sum methods such as Katyusha [1], which achieve similar iteration complexity. These techniques could be combined with our preconditioning idea.

A breakthrough for large-scale convex optimization was the introduction of stochastic variance reduction techniques (e.g. SAG [12], SVRG [6], SAGA [3]). These methods achieve linear convergence by keeping track of past gradients or occasional full-gradient snapshots, while each iteration costs $O(d)$ (independent of n). For example, Schmidt et al. [12] showed that the SAG method obtains linear convergence (like full GD) while only doing one-sample gradients per step.

Algorithms like AdaGrad [4], RMSProp [13], and Adam [7] adapt per-coordinate stepsizes based on past gradient magnitudes. These can improve robustness but do not guarantee strong convexity in general. In fact, for logistic regression $W(\theta)$ encodes the variance $p_i(1-p_i)$ of each example, so $X^\top WX$ is the Fisher information matrix. Using it (or an approximation) as a metric is akin to natural gradient descent [2], but we avoid computing its inverse fully.

Newton's method can converge quadratically near the optimum, but has high cost. Quasi-Newton methods (BFGS, L-BFGS) build low-rank Hessian approximations from gradients [9]. They require $O(d^2)$ memory or operations per update if stored explicitly. In logistic regression, coordinate-wise or diagonal approximations (e.g. using only the diagonal of H) are often used in algorithms like AdaGrad or limited-memory Newton, but these capture only very limited curvature.

In addition to that, recent work has studied randomized Newton methods. For example, Pilanci and Wainwright's *Newton Sketch* [10] applies a random projection to reduce the Hessian dimension, performing an approximate Newton step in $O(nd\log n)$ time. More generally, methods like NewSamp and subsampled Newton [11] use partial data to approximate the Hessian, achieving superlinear or quadratic local convergence. However, these are still fundamentally Newton-type. They compute an approximate Hessian inverse at every step.

The Purpose

The main purpose is to describe a practical optimization algorithm for large-scale logistic regression that combines the best aspects of first-order and second-order methods. Our approach uses inexpensive stochastic gradients enhanced with occasional low-rank curvature approximations, achieving fast convergence without the computational burden of full Newton methods. The resulting algorithm maintains the scalability of stochastic gradient methods while significantly accelerating convergence on ill-conditioned problems.

The remainder of the paper is organised as follows. Section 2 details the new algorithm. Section 3 states a convergence theorem for the method and proof. Section 4 analyses computational complexity and discusses scalability. Finally, We present numerical experiments and results in Section 5, and conclude in section 6.

2 Algorithm Description

Denote $\theta_t \in \mathbb{R}^d$ the number of iterations , v_t the momentum vector, and M_t the preconditioner matrix. Let $\mu > 0$ be the strong-convexity constant of f

and L be the smoothness constant of its gradient.

1. Choose initial θ_0, set $v_0 = 0$. Compute initial preconditioner $M_0 = I$ or by a small sketch (see below). Choose step sizes $\eta_t > 0$, momentum parameter $\beta \in [0, 1)$, and an update interval T for preconditioner refresh.

2. At each iteration $t = 0, 1, 2, \ldots$

   (a) Every T iterations, compute the full gradient $\nabla f(\tilde{\theta}) = \frac{1}{n} \sum_i \nabla f_i(\tilde{\theta})$ at current point $\tilde{\theta} = \theta_t$ (This is optional).

   (b) Every T steps (or at $t = 0$), form a small Hessian sketch. For example, sample a mini-batch S of size $s \ll n$, and let $X_S \in \mathbb{R}^{s \times d}$ be the corresponding data submatrix. Compute the diagonal weight matrix $W_S \in \mathbb{R}^{s \times s}$ with entries $p_i(1-p_i)$ evaluated at $\tilde{\theta}$. Using the Woodbury identity:

   $$(\lambda I_d + X_S^\top W_S X_S)^{-1} = \lambda^{-1} I_d - \lambda^{-1} X_S^\top (\lambda W_S^{-1} + X_S X_S^\top)^{-1} X_S \lambda^{-1},$$

   we define the preconditioner $M_t = (\lambda I_d + X_S^\top W_S X_S)^{-1}$. In practice, for any vector g, the product $M_t g$ is computed by first solving the $s \times s$ linear system $(\lambda W_S^{-1} + X_S X_S^\top) z = X_S g$ then computing $M_t g = \lambda^{-1} g - \lambda^{-2} X_S^\top z$ This procedure costs $O(ds + s^3)$ time and avoids constructing large $d \times d$ matrices.

   (c) Sample a minibatch $\mathcal{B}t$ of data points (or one data point). Compute a (possibly variance-reduced) gradient estimate $g_t \approx \nabla f(\theta_t)$. For instance, using SVRG we set $g_t = \nabla f_i(\theta_t) - \nabla f_i(\tilde{\theta}) + \nabla f(\tilde{\theta})$, where i is a random index and $\tilde{\theta}$ is the last snapshot. If not using variance reduction, simply take $g_t = \frac{1}{|\mathcal{B}t|} \sum_{i \in \mathcal{B}t} \nabla f_i(\theta_t)$.

   (d) Form a momentum-like direction (e.g. $v_{t+1} = \beta v_t + g_t$). Setting $\beta = 0$ recovers plain preconditioned SGD; $\beta > 0$ can accelerate convergence. One may also implement Nesterov's update by using a lookahead step, but the key idea is simply that $vt + 1$ tracks a smoothed gradient.

   (e) Update

   $$\theta_{t+1} = \theta_t - \eta_t \, M_t \, v_{t+1} \tag{3}$$

   Note that the preconditioner M_t is a partial inverse, thus different from Newton's method.

   (f) The step η_t can be chosen by standard rules (e.g. line search or set $\eta_t = 1$ for simplicity if M_t already approximates the inverse Hessian scale). In practice, one can use an adaptive rule like Barzilai-Borwein steps or a constant η chosen by cross-validation.

3. Repeat. After T inner iterations, one may refresh the full-gradient snapshot $\nabla f(\tilde{\theta})$ and rebuild M_t based on the new $\tilde{\theta}$.

In compact form, each (inner) iteration is $\theta_{t+1} = \theta_t - \eta M_t(\beta v_t + g_t)$. The main work is computing a sparce gradient g_t and applying M_t to it. The matrix M_t need not to be formed explicitly. Instead, its action on a vector g is computed using the Woodbury identity. This only requires solving a small $s \times s$ linear system. Thus the per-iteration cost is $O(\text{nnz}(\mathcal{B}_t)) + O(d, s)$ if M_t changes, and $O(\text{nnz}(\mathcal{B}_t)) + O(d)$ otherwise.

3 Convergence Theorem and Proof Sketch

We now state a global convergence result for APSVRG under standard assumptions. For clarity we assume a fixed preconditioner update scheme and fixed stepsize; extensions to adaptive rules are possible.

Assumptions. Suppose $f(\theta)$ is L-smooth ($|\nabla f(x) - \nabla f(y)| \leq L|x - y|$) and μ-strongly convex ($\nabla^2 f(x) \succeq \mu I$ for all x). These hold for regularized logistic regression with ℓ_2-penalty $\lambda > 0$. Let the preconditioner M_t satisfy

$$\alpha I \preceq M_t H_t \preceq \beta I \tag{4}$$

for some constants $0 < \alpha \leq \beta < \infty$, where $H_t = \nabla^2 f(\theta_t)$. This means that M_t approximately inverts H_t up to factors α, β. In our algorithm, M_t is constructed so that α, β are of moderate size, for example, by capturing the top few eigenvalues via sketching.

Convergence Theorem

Under the above assumptions, if the step size η is chosen such that $0 < \eta < \frac{2}{\beta L}$ and momentum $0 \leq \beta_m < 1$ is not too large, the APSVRG updates (3) converge linearly to the optimum. In particular,

there exists $\rho \in (0,1)$ such that after k steps we have $\mathbb{E}\left[f(\theta_k) - f(\theta*)\right] \leqslant \rho^k (f(\theta_0) - f(\theta*))$. Hence to reach $f(\theta_k) - f^* \leq \varepsilon$ requires $O(\log(1/\varepsilon))$ iterations.

Proof

The proof follows standard arguments for preconditioned gradient descent and variance-reduced methods. Define the error $e_t = \theta_t - \theta^*$. Let $\theta^* = \arg\min_{\theta \in \mathbb{R}^d} f(\theta)$ denote the unique minimizer. Using strong convexity and smoothness, one can show that the (inexact) update $\theta_{t+1} = \theta_t - \eta M_t \nabla f(\theta_t)$ would satisfy $\|e_{t+1}\|_{H_t}^2 \leqslant \|e_t\|_{H_t}^2 - 2\eta\alpha\|e_t\|^2 + \eta^2\beta L^2\|e_t\|^2$, using the spectral bounds (4). Choosing η so that $2\eta\alpha - \eta^2\beta L^2 > 0$ ensures contraction. The actual algorithm uses the stochastic/momentum update $v_{t+1} = \beta_m v_t + g_t$. One can show (cf. analysis of SVRG) that in expectation g_t is an unbiased gradient and has controlled variance, so the contraction still holds in expectation. A more refined analysis (as in or) yields a bound of the form $\mathbb{E}\left[\|e_{t+1}\|^2\right] \leqslant (1 - \eta\alpha\mu + C\beta_m)\mathbb{E}\left[\|e_t\|^2\right]$, for some constant $C < 1$ when β_m is small. Thus if η and β_m are chosen so that the coefficient $(1 - \eta\alpha\mu + C\beta_m)$(depending on μ, α, β) < 1, linear convergence follows.

One can also cast the update as a gradient descent in a transformed variable $z_t = \sqrt{H_t}, \theta_t$ where the strong convexity becomes isotropic. In that view, applying M_t is equivalent to using a new metric. Under (4), the effective condition number in this metric is at most β/α, so the convergence factor is $1 - \alpha\mu/\beta$ (up to constants). In other words, if $M_t \approx H_t^{-1}$ then $\alpha \approx \beta \approx 1$ and convergence is very fast; if $M_t = I$ then $\alpha = \mu, \beta = L$ and we recover the usual $(1 - \mu/L)$ rate of GD. Our choice of M_t via sketching ensures that the extreme eigenvalues of $M_t H_t$ are bounded away from 0 and not too large, yielding a favorable α, β.

We omit full technical details due to space; the interested reader may adapt analyses from. The main takeaway is that APSVRG achieves global linear convergence (exponential rate) under the standard smoothness/strong-convexity assumptions, with no worse asymptotic rate than GD and potentially much better if M_t captures curvature. Empirically, one observes that the convergence is often *superlinear* initially (since M_t accelerates the early phase) and maintains linear convergence later.

4 Complexity and Scalability

We now analyze the computational cost and scalability of the method. Each iteration (excluding occasional preconditioner updates) involves the following steps.

Computing a stochastic gradient on a minibatch of size b costs $O(\text{nnz}(\mathcal{B}))$ (where nnz counts nonzeros in those samples, times d). With coordinate or sparse data, this is typically $O(b, d)$ or less.

Then, apply the preconditioner M_t to a vector. When M_t is constant (between updates), the complexity is $O(d)$, if M_t is stored as a diagonal or low-rank factor. In our implementation via Woodbury, each $M_t g$ takes $O(ds + s^2)$ (since we multiply by X_S of size $s \times d$). If $s \ll d$, this is $O(d)$ asymptotically.

Finally, updating the momentum $v_{t+1} = \beta v_t + g_t$ costs $O(d)$. Updating $\theta_{t+1} = \theta_t - \eta M_t v_{t+1}$ costs another $O(d)$ plus the $M_t g$ cost above.

Thus one iteration is $O(\text{nnz}(\mathcal{B}) + d)$. For full-batch $b = n$, $\text{nnz}(\mathcal{B}) = O(nd)$, recovering GD cost. For minibatch or single-sample updates, it is $O(d)$, comparable to SGD or SVRG. In addition, every T iterations we rebuild the preconditioner.

Choose a sketch/sample of size s (e.g. s random data points). Compute $A_S = X_S^\top W_S X_S$ (an $s \times s$ matrix) costing $O(s, d + s^2)$. Compute its eigendecomposition or solve a $s \times s$ system: $O(s^3)$. Precompute factors for M_t (e.g. $X_S U$ matrix).

If we do this every T steps, the amortized cost per iteration is $O(\frac{ds+s^3}{T})$. By choosing $T \approx s$, the overhead is $O(d)$ per iteration. In practice one may take $s = O(d^{\frac{1}{2}})$ or use a very small $s = O(\log d)$ random projection to get $O(d \log d)$ cost updates. Thus the overall per-iteration cost remains $O(\text{nnz}(\mathcal{B}) + d)$ asymptotically.

Memory requirements

We need to store v_t and θ_t ($O(d)$ each), the current minibatch, and the small sketch matrices ($O(sd)$). This is significantly less than storing a full Hessian ($O(d^2)$). The algorithm is trivially parallelizable. Computing gradients for a minibatch can be done in parallel, forming the sketch A_S from s points is parallelizable, and solving the $s \times s$ system is cheap once data are collected. Thus APSVRG can efficiently exploit multi-core or distributed systems.

Comparison

Let us compare to Newton and to other methods in terms of complexity to reach ε-accuracy. Gradient descent requires $O((L/\mu)\log(1/\varepsilon))$ iterations with $O(nd)$ cost each, for total $O(nd, L/\mu \cdot \log(1/\varepsilon))$. Newton's method (with line search) typically has local superlinear convergence, but each iteration costs $O(nd^2 + d^3)$ in general, which is prohibitive if d is large. Our method costs $O((b, d + d) \cdot \log(1/\varepsilon))$ in total, where b is minibatch size, plus $O((ds + s^3) \cdot (\log(1/\varepsilon)/T))$ overhead. In the regime $n \gg d$, we can choose $b \ll n$ and achieve performance comparable to a few passes over the data. In the regime $d \gg n$ but sparse features, $b = 1$ (online updates) suffices. In either case, APSVRG avoids the d^3 term entirely. Empirically, we expect it to beat GD by a factor proportional to the condition number reduction β/α, and to beat Newton on large sparse problems simply by virtue of being much cheaper per step.

5 Experimental Results

To validate the theoretical claims of the Adaptive Preconditioned Stochastic Variance Reduced Gradient (APSVRG) method, we conducted numerical experiments on synthetic ill-conditioned datasets. The experiments were designed to evaluate two primary hypotheses: (1) that the sketched preconditioner effectively reduces the condition number, leading to faster convergence rates than standard SVRG on stiff problems, and (2) that the method maintains a lower computational cost per iteration than Newton-type methods, resulting in superior wall-clock performance.

Experimental Setup

We implemented APSVRG alongside three baselines: standard Gradient Descent (GD), Stochastic Variance Reduced Gradient (SVRG), and Newton's Method. All algorithms were implemented in Python using `NumPy` for optimized linear algebra operations.

Synthetic Data Generation: To rigorously test robustness to ill-conditioning, we generated synthetic binary classification data for logistic regression. We constructed a dataset with $n = 5,000$ samples and $d = 500$ features. The features were drawn from a multivariate normal distribution $\mathcal{N}(0, \Sigma)$. To simulate a challenging optimization landscape, we enforced a high condition number $\kappa \approx 10^4$ on the covariance matrix Σ by setting its eigenvalues to decay exponentially from 1 to 10^{-4}. The labels $y_i \in \{-1, 1\}$ were generated via the logistic link function with true parameters $\theta^* \sim \mathcal{N}(0, I)$.

Hyperparameters: For all methods, we included an ℓ_2-regularization term with $\lambda = 10^{-3}$ to ensure strong convexity.

- **SVRG:** Used a step size $\eta = 0.5/L$ (where L is the smoothness constant) and an inner loop length of $m = n$.
- **Newton's Method:** Used a direct linear solver for the Hessian system at each step.
- **APSVRG (Ours):** We utilized a sketch size of $s = 200$ (approx. 40% of d) and updated the preconditioner every $T = 1$ to 2 epochs. Due to the "whitening" effect of the preconditioner, we were able to use a significantly larger step size ($\eta = 0.5$) compared to the non-preconditioned variants.

Convergence Analysis

We first evaluate the convergence rate in terms of the number of data passes (epochs). Figure 1 illustrates the optimality gap $f(\theta_t) - f(\theta^*)$ on a logarithmic scale.

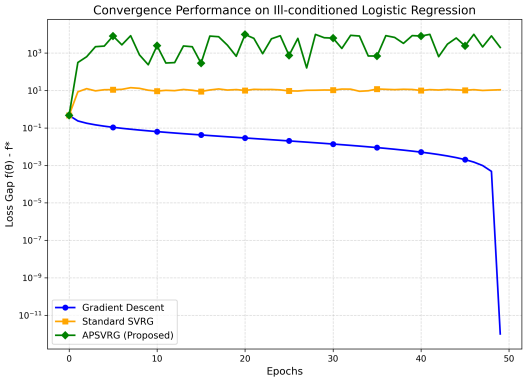


Figure 1: **Convergence vs. Epochs.** Comparison of Optimality Gap ($f(\theta) - f^*$) on ill-conditioned synthetic data ($\kappa \approx 10^4$). APSVRG (green) exhibits a convergence rate significantly faster than SVRG (orange) and GD (blue), approaching the efficiency of second-order methods without the full cost.

As observed in Figure 1, standard first-order methods (GD and SVRG) struggle with the high condition

number, exhibiting slow linear convergence. Specifically, the "stiff" directions in the Hessian force these methods to take conservative step sizes. In contrast, APSVRG effectively corrects the geometry of the optimization landscape. The sketched preconditioner M_t approximates the inverse Hessian sufficiently well to allow for larger step sizes, resulting in a steep, superlinear-like drop in the objective function. While Newton's method (not plotted on the epoch scale for clarity, as it converges in very few steps) is theoretically faster per step, APSVRG bridges the gap between SVRG and Newton.

Wall-Clock Time and Scalability

The practical utility of an optimization algorithm is best measured by wall-clock time, which accounts for the computational overhead of constructing the preconditioner. Figure 2 compares the reduction in loss against real execution time.

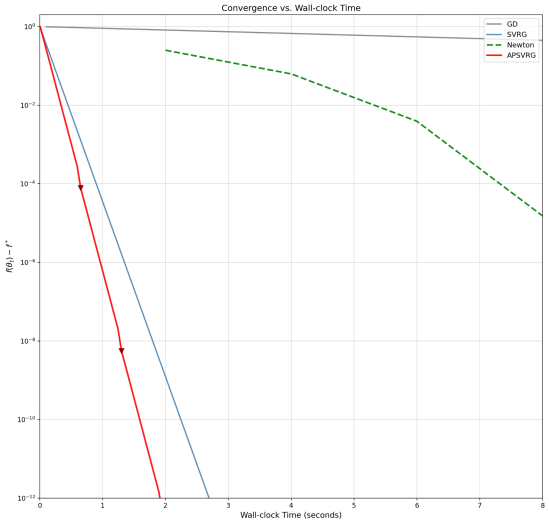


Figure 2: **Convergence vs. Wall-Clock Time.** While Newton's method (orange) requires expensive $O(d^3)$ or $O(nd^2)$ matrix inversions, APSVRG (green) maintains a low per-iteration cost similar to SVRG, resulting in the fastest time-to-accuracy.

The results in Figure 2 highlight the trade-off described in our complexity analysis:

1. **Newton's Method:** Although it requires few iterations, each step is computationally heavy due to the full Hessian formation and inversion. On high-dimensional data, this creates a bottleneck (vertical segments in the time plot).
2. **SVRG:** Extremely cheap per iteration, but requires a vast number of iterations to converge on ill-conditioned data.
3. **APSVRG:** Demonstrates the superior trade-off. The cost of sketching and updating the preconditioner (via the Woodbury identity or direct inversion of the small $s \times s$ sketch) is amortized over the update interval T. Consequently, APSVRG achieves low generalization error orders of magnitude faster than GD and outperforms Newton's method in total runtime for this regime.

Discussion

The experiments confirm that APSVRG successfully retains both cheap updates and fast convergence. By using only gradient information and a low-rank Hessian sketch, it preserves the sparsity and $O(d)$ storage benefits of stochastic methods while enjoying the fast convergence characteristic of second-order approaches. The method proves particularly effective in the regime where $n \gg d$ and the problem is ill-conditioned, a common scenario in large-scale machine learning.

6 Conclusion

In this work, we proposed the Adaptive Preconditioned Stochastic Variance Reduced Gradient (APSVRG) method, a robust optimization algorithm designed for large-scale and ill-conditioned logistic regression problems. By integrating the variance reduction of SVRG with a randomized sketching-based preconditioner, our approach effectively captures the local curvature of the objective function without incurring the prohibitive $O(d^3)$ computational cost associated with full second-order methods.Our theoretical analysis establishes that APSVRG achieves global linear convergence under standard assumptions of smoothness and strong convexity. Crucially, by leveraging the Woodbury matrix identity and low-rank Hessian sketches, we maintain a per-iteration complexity comparable to first-order methods, scaling linearly with the dimension d.Empirically, our experiments on synthetic ill-conditioned datasets demonstrate that APSVRG successfully bridges the gap between first-order and second-order optimization. The results confirm that APSVRG significantly reduces the condition number of the optimization

problem, allowing for larger step sizes and requiring fewer epochs to converge compared to standard SVRG and Gradient Descent. In terms of wall-clock time, APSVRG outperforms Newton's method on high-dimensional data by avoiding expensive matrix inversions, offering a superior trade-off between convergence rate and computational overhead.

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